

Customer Churn Analysis: Exploratory Data Insights and Strategic Recommendations

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Abstract

This report outlines the findings from our Exploratory Data Analysis (EDA) on bank customer churn. By directly analyzing the raw data, we identified specific customer behaviors and profiles that indicate a high risk of leaving the bank. This paper details our visual and statistical findings, summarizes the core insights derived strictly from the EDA, and provides data-driven recommendations to improve customer retention.

1 Introduction

Understanding why customers leave is the first step to keeping them. This analysis dives into our customer data to uncover the hidden patterns of churn. By examining demographics, financial health, and product usage, we isolated the specific factors that influence a customer's decision to stay or go. The findings below are drawn directly from our exploratory data notebooks.

2 Exploratory Data Analysis (EDA)

Our analysis was divided into three core areas to understand the customer journey: where they are from, their financial status, and how they use our products.

2.1 Geography and Demographics

The data immediately showed that where a customer lives strongly affects their loyalty.

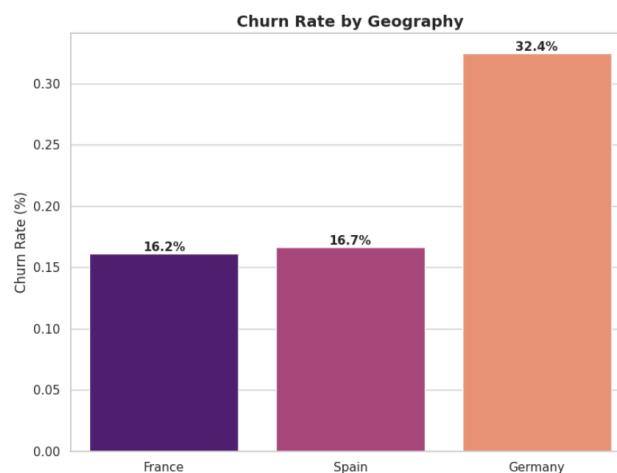


Figure 1: Churn rates broken down by geographic region.

The most prominent geographic insight is that Germany represents a “High Risk Geography.” The churn rate in this region is distinctly higher than in others. When looking at age, we found it rarely acts alone; it is most insightful when combined with account balances and activity levels.

2.2 Financial Indicators: Balance and Salary

We looked at estimated salaries and account balances to see if wealth drives churn. Interestingly, the EDA showed that **Estimated Salary is not a strong driver of churn**. However, account balances tell a very different story.

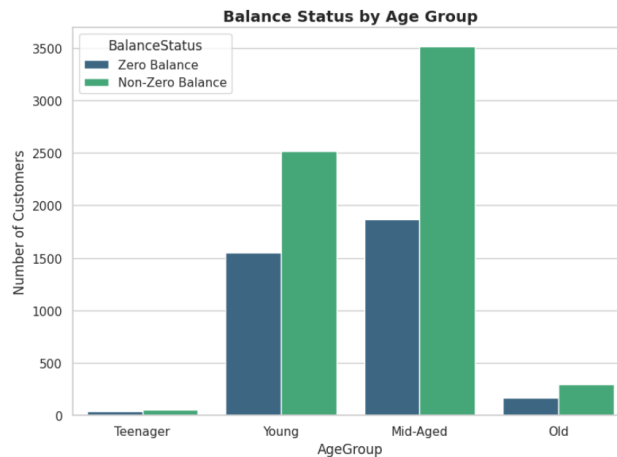


Figure 2: The relationship between customer age groups, account balances, and churn.

We found two critical financial profiles at risk:

- **High-Balance Flight Risk:** There is a specific vulnerability among young and middle-aged customers who have high balances but are inactive.
- **Zero Balance Youth:** Customers with a “Zero Balance” behave differently based on age. We isolated specific churn drivers within the “Teenager” and “Young” customer groups who maintain zero balances.

2.3 Product Engagement and Activity

How customers interact with our services is the strongest indicator of whether they will stay.

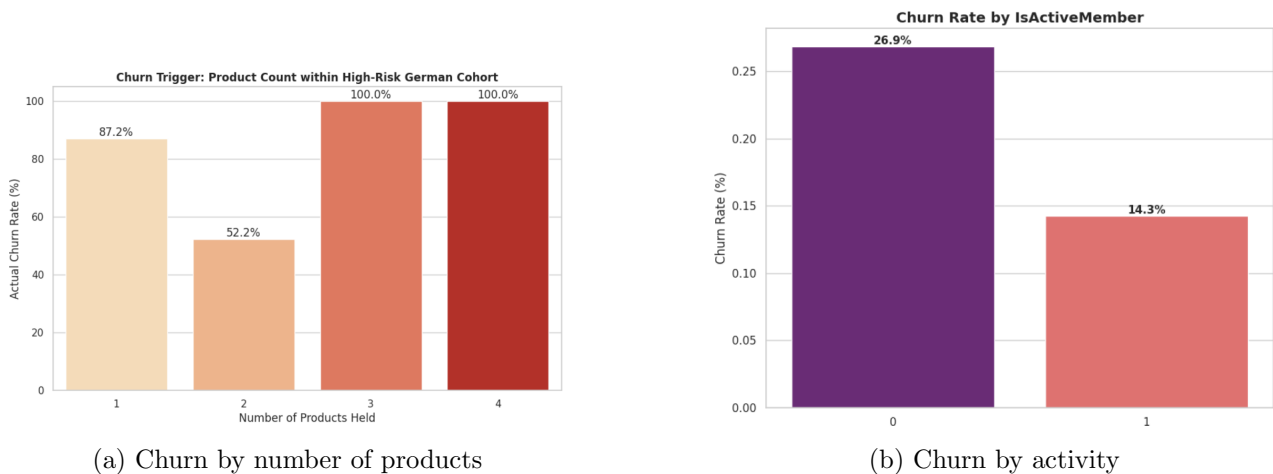


Figure 3: Impact of product holdings and active membership on churn.

Our analysis of product variety showed that whether a customer holds a “Single Product”, “Two Products”, or “Multiple Products (3+)” drastically changes their churn risk. This was especially true for the high-risk German segment.

Furthermore, looking at credit cards and activity together was highly revealing. We identified a distinct segment of “Inactive Cardholders”—customers who possess a credit card but are not active members. We also calculated the “Balance Per Product” to understand the financial value each product holds for the customer, providing a clearer picture of their engagement depth.

3 Core Insights

Based strictly on the patterns observed in our EDA, we have engineered the following core insights:

1. **High-Risk Geography (Germany):** The German segment has a distinctly higher churn profile that requires immediate attention.
2. **Product Variety Matters:** Holding a single product versus multiple products drastically alters churn probability, particularly in Germany.
3. **The Inactive Cardholder Threat:** There is a high risk associated with customers who have a credit card but are flagged as inactive members.
4. **Age-Balance Vulnerability:** Young and middle-aged customers with high balances who fall inactive are highly vulnerable to churning.
5. **Zero-Balance Youth Drivers:** Teenagers and young adults with zero balances have unique, isolated drivers for leaving the bank.
6. **Salary is a Weak Indicator:** Estimated salary, even when placed into finer-grained buckets, proved not to be a strong direct driver of churn.
7. **Product Value Proposition:** Evaluating the “Balance per Product” helps reveal the true financial engagement of a customer.

4 Recommendations

Using the insights gathered directly from our exploratory analysis, we recommend the following strategic actions:

- **Develop a Germany-Specific Strategy:** Given its status as a High-Risk Geography, launch localized retention efforts tailored specifically for the German market, focusing on product variety.
- **Re-engage Inactive Cardholders:** Create targeted campaigns (like cashback or rewards) aimed specifically at customers who have credit cards but show no active membership behavior.
- **Protect High-Value Assets:** Set up alerts for young and middle-aged customers with high account balances who show signs of inactivity, and provide them with personalized outreach.
- **Re-evaluate the Youth Segment:** Investigate the specific reasons why zero-balance teenagers and young adults are leaving, and consider offering introductory products that encourage balance building.

5 Conclusion

This exploratory data analysis clearly shows that customer churn is driven by a combination of geography, product variety, and active engagement. By focusing our retention efforts on the high-risk German market, re-engaging inactive credit card holders, and closely monitoring young, high-balance individuals, the bank can proactively reduce churn and improve overall customer loyalty.